

Current Multimedia, GUI & Media Formats

Lesson 65 — Multimedia, GUI and File-Format Recall

JKSSB Computer Awareness

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What Is Multimedia?

- ▶ **Multimedia** is content that combines more than one medium in a single presentation.
- ▶ The media may be **text, graphics, audio, video**, and **animation**.
- ▶ Multimedia is stored in digital form, so each medium must first be captured and converted into binary data.
- ▶ Examples: a training video with narration, a web page with pictures, and a presentation with sound.

The one idea

Multimedia = multiple media types presented together, all reduced to digital form the computer can process.

The Five Building Blocks of Multimedia

Element	Meaning
Text	Readable characters; the base medium.
Graphics	Still images, drawings, and photographs.
Audio	Sound, music, and speech.
Video	Moving pictures, usually with sound.
Animation	A sequence of images that creates motion.

Text and graphics are static; audio, video, and animation are time-based media.

Digitising Media: Sampling

- ▶ Analogue sound and light are **continuous** signals; a computer stores only discrete numbers.
- ▶ **Sampling** means measuring the signal at regular intervals and recording each measurement.
- ▶ The **sampling rate** is the number of samples taken per second, measured in hertz (Hz).
- ▶ A **higher** sampling rate gives better quality but a larger file.

Sampling answers two questions

How often do we measure the signal, and how many measurements do we store each second?

Quantization and Bit Depth

- ▶ Each sample is a precise value; the computer must round it to the nearest allowed level.
- ▶ **Quantization** is the process of mapping each sample to one of a fixed set of discrete levels.
- ▶ The number of levels is set by the **bit depth** of each sample.
- ▶ **More levels** (higher bit depth) means less rounding error and better quality, but a larger file.

Trap alert

Sampling = how often we measure. Quantization = how precisely each measurement is recorded. Do not confuse the two.

Compression: Lossy vs Lossless

- ▶ **Compression** reduces the size of a media file.
- ▶ **Lossless** compression restores the original exactly — no data is lost (example: PNG, GIF, ZIP).
- ▶ **Lossy** compression discards some data to shrink the file further (examples: JPEG, MP3).
- ▶ Lossy files are usually smaller; lossless files preserve full quality.

Which to choose

Use lossless when exactness matters; use lossy when a smaller file matters more than perfect fidelity.

Exam point

JPEG, MP3, and MP4 are lossy formats; PNG and GIF are lossless.

Synchronization of Media Streams

- ▶ In a multimedia file, audio and video are separate **streams**.
- ▶ **Synchronization** keeps those streams aligned in time.
- ▶ If they drift apart, speech no longer matches the moving lips — a **sync** error.
- ▶ Synchronization is what makes a video play with the sound in the correct place.

Simple view

Synchronization means the picture and the sound stay together at every moment of playback.

GUI vs CLI

- ▶ A **GUI** (Graphical User Interface) lets the user work with **pictures, icons, and menus** instead of typed commands.
- ▶ A **CLI** (Command-Line Interface) accepts typed text commands.
- ▶ The GUI is easier for beginners; the CLI is often faster for experts.
- ▶ Windows and modern Linux desktops use a GUI.

Trap alert

GUI = **Graphical** User Interface. It is not “General” or “Graphic” alone; the exact full form is asked directly.

GUI Elements: WIMP

- ▶ The four classic GUI elements are remembered as **WIMP**: **W**indows, **I**cons, **M**enus, and **P**ointers.
- ▶ **Window**: a rectangular area that holds a program or document.
- ▶ **Icon**: a small picture that represents a file, folder, or program.
- ▶ **Menu**: a list of commands offered by the software.
- ▶ **Pointer**: the on-screen arrow moved by the mouse or touchpad.

WIMP full form

Windows, Icons, Menus, Pointers — the standard interaction set of a GUI.

Dialogue Boxes and WYSIWYG

- ▶ A **dialogue box** is a small window that asks for input or confirms an action (for example, Save As or Print).
- ▶ It usually contains text boxes, drop-down lists, check boxes, radio buttons, and **OK / Cancel** buttons.
- ▶ **WYSIWYG** (What You See Is What You Get) means the screen shows the document exactly as it will be printed.

Worked example

In MS Word, the page on screen looks like the printed page. That is WYSIWYG. When the Print dialogue box opens, the settings appear inside a dialogue box.

Image Formats: PNG, JPEG, GIF

Format	Full form	Nature and typical use
PNG	Portable Network Graphics	Lossless; supports transparency; logos, web graphics.
JPEG	Joint Photographic Experts Group	Lossy; best for photographs.
GIF	Graphics Interchange Format	Lossless but limited to 256 colours; simple animation.

GIF is an image format that also supports short, simple animation.

Choosing the Right Image Format

- ▶ **JPEG** for photographs and complex pictures — small size, some quality loss.
- ▶ **PNG** when transparency or exact colours are needed — lossless.
- ▶ **GIF** for simple graphics and small animations — limited to 256 colours.
- ▶ **BMP** is a large, uncompressed bitmap format, not a web format.

Trap alert

JPEG is **lossy**, PNG and GIF are **lossless**. Neither PNG nor JPEG supports animation; only GIF does among the three.

Audio Format: MP3

- ▶ **MP3** stands for **MPEG-1 Audio Layer III** (also written MPEG Audio Layer III).
- ▶ It is a **lossy** audio format: it removes sound data the ear is unlikely to notice.
- ▶ Its small size made it the standard format for portable music and downloads.
- ▶ **WAV** is an uncompressed audio format; MP3 is compressed.

Keep the family straight

MPEG is the Moving Picture Experts Group; MP3 is its audio layer, MP4 is its later container.

Video Format: MP4

- ▶ **MP4** stands for **MPEG-4 Part 14**.
- ▶ It is a **container** format: it can hold video, audio, and subtitles in one file.
- ▶ It is widely used for streaming, downloads, and mobile video.
- ▶ **AVI** (Audio Video Interleave) is an older video container.

Container vs codec

MP4 is the container (the box); the codec inside it actually compresses the video and audio.

Media Format Quick Table

Format	Type	Compression	Best use
PNG	Image	Lossless	Web graphics, transparency
JPEG	Image	Lossy	Photographs
GIF	Image	Lossless	Simple graphics, animation
MP3	Audio	Lossy	Music and speech
MP4	Video	Lossy	Video with sound

Full forms are high-yield: learn every expansion exactly.

Plotter

- ▶ A **plotter** is an **output** device that draws on paper.
- ▶ It produces **vector graphics** — lines and curves defined by coordinates.
- ▶ It is used for **large engineering drawings**, maps, blueprints, and banners.
- ▶ Pen plotters use pens; modern inkjet plotters spray ink.

Trap alert

A plotter is an **output** device, not input. It is chosen for large technical or engineering drawings, not for ordinary text. (PYQ-29)

Impact vs Non-Impact Printers (Revisit)

Point	Impact	Non-impact
Action	Strikes the paper	No striking
Examples	Dot-matrix, daisy-wheel, line printer	Inkjet, laser, thermal
Copies	Carbon copies possible	No carbon copies
Noise	Noisy	Quiet

Both are output devices; the difference is whether the mechanism strikes the paper.

Full Forms You Must Recall

Short	Full form	Note
GUI	Graphical User Interface	Icons and menus, not typed commands
CLI	Command-Line Interface	Typed commands
WYSIWYG	What You See Is What You Get	Screen matches print
WIMP	Windows, Icons, Menus, Pointers	Classic GUI elements
MP3	MPEG-1 Audio Layer III	Lossy audio
MP4	MPEG-4 Part 14	Video container

Trap Alert: Format Facts to Keep Straight

- ▶ **Sampling** = how often we measure; **quantization** = how precisely each sample is stored.
- ▶ **JPEG, MP3, MP4** are lossy; **PNG, GIF** are lossless.
- ▶ **MP3** is audio; **MP4** is video; do not swap their full forms.
- ▶ A **plotter** is an output device for large drawings.
- ▶ **Impact** printers strike the paper; **non-impact** printers do not.
- ▶ A **dialogue box** asks for input; **WYSIWYG** shows the print layout on screen.

Lesson 65: Recall

- ▶ Multimedia combines text, graphics, audio, video, and animation.
- ▶ **Sampling** measures a continuous signal; **quantization** maps each sample to discrete levels; **bit depth** sets the number of levels.
- ▶ **Lossless** restores the original (PNG, GIF); **lossy** discards data (JPEG, MP3).
- ▶ **Synchronization** keeps audio and video lined up in time.
- ▶ GUI elements are **Windows, Icons, Menus, Pointers** (WIMP); **WYSIWYG** matches screen to print.
- ▶ **PNG, JPEG, GIF** are image formats; **MP3** is audio; **MP4** is video.
- ▶ A **plotter** is an output device for large drawings; **impact** printers strike, **non-impact** printers do not.